INDUSTRIAL TRAINING PROJECT REPORT

PROJECT PILOT

[A PROJECT MANAGEMENT WEBSITE]

Submitted in Partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

SESSION: (2023-2027)

Degree of Bachelor of Technology Computer Science Engineering

Submitted By:
Alisha
Roll no: -2326268

Department of Computer Science & Engineering
MODERN GROUP OF COLLEGES, PANDORI BHAGAT, MUKERIAN(PUNJAB)
Affiliated To
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SUBMITTED TO: -
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DECLARATION

I here by declare that the Project Report entitled "**PROJECT PILOT**" is an authentic record of my work as requirements of project during the period of **SIXTH semester** for the award of degree of **B. Tech (CSE), Modern Group Of Colleges, Pandori Bhagat (Mukerian)** under the guidance of **Miss Janvee Sharma.**

Date: _____

ALISHA (2326268)

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

Signatures

Examined by:

1. 2. 3. 4.

Head of Department
(Signature and Seal)

ACKNOWLEDGEMENT

I express my sincere gratitude to the I.K. Gujral Punjab Technical University, Kapurthala for giving me the opportunity to work on the Major Project during my Sixth Semester of B. Tech (CSE) is an important aspect in the field of engineering.

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Alisha (2326268)

B. Tech (CSE) 6th Sem

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CHAPTER 1: INTRODUCTION

1.1: INTRODUCTION OF THE PROJECT

This pilot project has been launched to test a new Employee Project Management System designed to improve how employee tasks and projects are assigned, tracked, and completed. The system aims to support better planning, collaboration, and performance management within teams. this pilot project is focused on implementing and evaluating a new process aimed at improving employee engagement, performance tracking, and overall project efficiency within the organization. The feedback collected from this pilot will help us understand what works well and what needs to be improved. Based on the results, decisions will be made about adjusting the system or moving forward with a full implementation across the organization.

The purpose of the pilot is to introduce the system on a small scale — involving a selected group of employees or departments — to see how well it works in real situations. This initiative will help determine the effectiveness of the proposed system in:

- Enhancing transparency in task allocation and tracking.
- Encouraging better communication and collaboration among team members.
- Providing real-time insights into employee performance and project status.
- Streamlining reporting and feedback mechanisms. The pilot will run over a defined period, with active feedback collection from both managers and employees to ensure that the implementation meets organizational goals and user needs effectively.
- Tracking progress of tasks and projects in real time.
- Improving communication between team members and managers.
- Generating reports on employee performance and project timelines.
- Assigning work clearly to the right team members.

The Project Pilot is a preliminary implementation of a comprehensive employee management system designed to streamline the workflow between employees and administrators.

1.2: OBJECTIVE OF THE PROJECT

The primary objective of a project pilot for both employees and administrators is to test the feasibility and effectiveness of a new project, process, or system in a controlled environment before a wider rollout. This allows for the identification of potential issues, refinement of strategies, and gathering of valuable feedback, ultimately leading to a smoother and more successful implementation on a larger scale. The main objectives of the pilot project are:

- To test the effectiveness of the new employee project management system in a real work environment.
- To evaluate how well tasks and projects can be assigned, monitored, and completed using the new system.
- To assess user experience and ease of use for both employees and managers interacting with the system.
- To identify potential technical or operational issues before full-scale implementation.
- To gather feedback from participants to make necessary improvements and ensure the system meets organizational needs.
- To measure improvements in productivity, communication, and performance tracking compared to existing processes.
- To determine scalability of the system across other teams, departments, or the entire organization**.**
- To introduce the system to a small group of employees and observe how it performs in real work situations.
- To understand how well employees and managers can use the system to manage daily tasks and project activities.
- To identify any problems or areas for improvement in the system before a full rollout.
- To test whether the system helps improve team coordination, productivity, and reporting.
- To collect useful feedback from users that can guide final adjustments.
- To evaluate if the system fits the organization's goals and workflow.
- Pilot projects provide data to assess the potential ROI of the project and make informed decisions about its future.

1.3: EXISTING SYSTEM

The current system used for managing employee projects and tasks is largely manual and fragmented. Most work is tracked through a combination of tools such as spreadsheets, email communication, and physical reporting. At present, the interaction between employees and the admin team is managed through manual processes such as emails, physical forms, verbal communication, or basic spreadsheets. There is no centralized digital system to streamline daily tasks, employee requests, or administrative approvals.

- For Employees:

- **Familiarization:**

  Employees can get hands-on experience with the new system, learning how to navigate it, access their information, and utilize its features.

- **Feedback:**

Employees can provide valuable feedback on the system's usability, identifying areas where it's intuitive or where it needs improvement.

- **Training:**

The pilot allows for targeted training to help employees effectively utilize the system.

- For Administrators:

- **System Functionality:**

Administrators can verify that the system meets the intended requirements and functions as expected.

- **Access Control:**

The pilot helps ensure that administrators can effectively manage user access and permissions within the system.

This approach leads to several challenges, including:

- Lack of real-time visibility into project progress and employee performance.
- Difficulty in tracking task ownership, deadlines, and status updates.
- Inefficient communication between team members and departments.
- Limited ability to generate accurate performance or progress reports.
- Higher chances of data errors, duplication, and miscommunication.

1.4: PROPOSED SYSTEM

The proposed system of Project Pilot is a centralized Employee and Admin Management Platform designed to streamline communication, task management, and workflow processes between employees and the administrative team. This digital solution aims to replace the current manual methods with a more efficient, transparent, and organized approach.

Key features of the proposed system include:

- **Task Assignment and Tracking:** Admins can assign tasks to employees, set deadlines, and track progress in real-time.
- **Request Management:** Employees can submit leave applications, resource requests, or queries directly through the platform.
- **Dashboard for Admin and Employees:** Separate dashboards provide quick access to tasks, status updates, and notifications.
- **Automated Notifications:** System-generated alerts for upcoming deadlines, new assignments, or approvals.
- **Performance Monitoring:** Tools for tracking employee participation, project contributions, and report generation.
- **Document Sharing:** A secure space for uploading and accessing important files or project-related document.
- **Success Criteria:** Establish what constitutes success for the pilot project. This could include user adoption rates, task completion times, or error reduction.
- **Timeline:** Set a realistic timeframe for the pilot, including when testing will begin, how long it will last, and when feedback will be collected.
- **Feedback Mechanisms:**

  Implement a system for collecting user feedback, such as surveys, interviews, or dedicated feedback channels. The proposed system will be tested in a selected department or group as part of this pilot project. The aim is to assess whether it improves:

- Workflow efficiency.
- Transparency between roles.
- Communication clarity.

1.5: PROJECT SCOPE

In-Scope Activities:

1.Employee Task Management

- Assigning tasks from admin to employees.
- Tracking task progress and completion status.
- Providing task history and deadlines.

2. Admin Dashboard Functions

- Viewing employee task reports.
- Managing employee requests (leaves, queries, etc.)
- Sending announcements and notifications.

3. Communication Tools

- Enabling two-way communication between employees and admin.
- System alerts for assignments, approvals, or changes

4. Performance Monitoring

- Generating reports on task completion.
- Reviewing employee responsiveness and engagement.

5. Feedback Collection

- Gathering user experience data from both admin and employees.
- Identifying usability issues or required improve me.

1.6: ADVANTAGES

1. Early Testing in a Real Environment

The pilot allows the organization to test the new system with actual admin and employee users, helping to observe how it performs in day-to-day operations.

2. Identifies Problems Before Full Rollout

By starting on a small scale, technical or workflow issues can be detected and resolved early, reducing the risk of large-scale failures.

3. Saves Time and Resources

Piloting the project prevents unnecessary investment in a system that might not work as expected. Adjustments can be made before spending on full implementation.

4. Gathers Valuable User Feedback

Direct feedback from both employees and admins helps improve system design, usability, and functionality based on real user needs.

DISADVANTAGES:

1.Limited Scope May Not Reflect Full Challenges

Since the pilot is run on a small scale, it might not expose all the issues that could occur during full implementation across the organization.

2. Resource and Time Consumption

Running a pilot requires time, planning, and effort from both employees and admins — which can temporarily affect regular work productivity.

3. Partial Results Can Be Misleading

The outcomes of a pilot may not fully represent what will happen in a larger setup, especially if the pilot group is not diverse or large enough.

CHAPTER 2: TOOLS & TECHNOLOGY

2.1: HARDWARE REQUIREMENTS

The system used to in order to create as well as test the project had the following specifications:

1. CPU – RYZEN5 3550H
2. GPU - Radeon RX560X
3. OS - Windows 11
4. SSD - 512GB
5. RAM - 8 GB

2.2: SOFTWARE REQUIREMENTS

Following software are used to build this project: -

1. Vs Code: - Visual Studio Code (VS Code) is a popular source code editor developed by Microsoft. It provides a wide range of features and extensions to support various programming languages and development workflows.

2.3: PROGRAMMING LANGUAGE: -

1. HTML:

HTML is an acronym which stands for Hyper Text Markup Language which is used for creating web pages and web applications. Let's see what is meant by Hypertext Markup Language, and Web page. It is explained as: -

Hyper Text: Hyper Text simply means "Text within Text." A text has a link within it, is a hypertext. Whenever you click on a link which brings you to a new webpage, you have clicked on a hypertext. Hyper Text is a way to link two or more web pages (HTML documents) with each other.

Markup language: A markup language is a computer language that is used to apply layout and formatting conventions to a text document. Markup language makes text more interactive and dynamic. It can turn text into images, tables, links, etc.

2.CSS (Cascading style sheet):

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colours, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple web pages to share formatting by specifying the relevant CSS in a separate .CSS file which reduces complexity and repetition in the structural content as well as enabling the .CSS file to be cached to improve the page load speed between the pages that share the file and its formatting. Separation of formatting and content also makes it feasible to present the same markup page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille-based tactile devices. CSS also has rules for alternate formatting if the content is accessed on a mobile device.

JavaScript (JS):

JavaScript (JS) is a lightweight, interpreted, or just-in-time compiled programming language with first-class functions. While it is most well-known as the scripting language for Web pages, many non-browser environments also use it, such as Node.js, Apache CouchDB and Adobe Acrobat. JavaScript is a prototype-based,

multi-paradigm, single-threaded, dynamic language, supporting object-oriented, imperative, and declarative (e.g. functional programming) styles. Read more about JavaScript.

This section is dedicated to the JavaScript language itself, and not the parts that are specific to Web pages or other host environments. For information about API specifics to Web pages, please see Web APIs and DOM.

JavaScript is used for interactive website.it is mainly used for:

1.Client-side validation

2.Dynamic drop-down menus

3.Displaying date and time

Bootstrap:

Bootstrap is an HTML, CSS and JS library that focuses on simplifying the development of informative web pages (as opposed to web applications). The primary purpose of adding it to a web project is to apply Bootstrap's choices of color, size, font and layout to that project. As such, the primary factor is whether the developers in charge find those choices to their liking. Once added to a project, Bootstrap provides basic style definitions for all HTML elements. The result is a uniform appearance for prose, tables and form elements across web browsers. In addition, developers can take advantage of CSS classes defined in Bootstrap to further customize the appearance of their contents. Bootstrap also comes with several JavaScript components which do not require other libraries like jQuery.

React:

React (also known as React.js or ReactJS) is a free and open-source front-end JavaScript library for building user interfaces based on components by Facebook Inc. It is maintained by Meta (formerly Facebook) and a community of individual developers and companies. React can be used to develop single-page, mobile, or server-rendered applications with frameworks like Next.js. Because React is only concerned with the user interface and rendering components to the DOM, react applications often rely on libraries for routing and other client-side functionality. A key advantage of React is that it only renders those parts of the page that have changed, avoiding unnecessary rendering of unchanged DOM elements. React code is made of entities called components. These components are modular and reusable. React applications typically consist of many layers of components. The components are rendered to a root element in the DOM using the React DOM library. When rendering a component, values are passed between components through props (short for "properties"). Values internal to a component are called its state.

Firebase:

Firebase is a set of backend cloud computing services and application development platforms provided by Google. It hosts databases, services, authentication, and integration for a variety of applications, including Android, iOS, JavaScript, Node.js, Java, Unity, PHP, and C++. Firebase is a comprehensive platform developed by Google for creating mobile and web applications. It provides a variety of tools and services that help developers build high-quality apps, grow their user base, and earn mode profit.

CHAPTER 3: FEASIBILITY

3.1: TECHNICAL FEASIBILITY

The project pilot is technically feasible based on available infrastructure, system requirements, ease of deployment, and compatibility with existing tools.

1.Compatibility with Existing Systems

The pilot system is designed to work alongside or integrate with current tools such as email, spreadsheets, or HR software. This ensures smooth data flow and avoids complete system replacement during the testing phase.

2. Ease of Implementation

The system can be hosted on a web platform (or internal server/cloud) and accessed by both employees and admin with minimal technical setup. Basic training and user guides are sufficient for most users to start using the system.

3. Data Security and Access Control

The system includes role-based access control — allowing admins to manage and monitor while employees can only access their relevant tasks and records. Standard data protection practices can be applied.

3.2: ECONOMIC FEASIBILITY

The economic feasibility of project pilot assesses whether the proposed employee-admin management system is cost-effective and financially justifiable during the pilot phase and potential full implementation**.**

1. Use of Existing Resources

The organization can leverage existing hardware (computers, servers, internet), reducing the need for new infrastructure or major investments.

2. Affordable Software and Tools

The proposed system is built using cost-effective technologies (e.g., web-based platforms, cloud storage) or may use open-source frameworks, which help reduce licensing fees or subscription costs during the pilot.

3. Cost Savings Through Automation

Once implemented, the system is expected to reduce the time and costs associated with manual task tracking, approvals, reporting, and internal communication between employees and admin staff.

3.3: OPERATIONAL FEASIBILITY

The operational feasibility evaluates whether the proposed employee-admin system can be successfully used and maintained by the people involved - including employees, admin staff, and IT support — as part of their daily operations.

1. User-Friendly Interface

The system is designed to be simple and intuitive, allowing both employees and admin staff to use it with minimal training. The interface supports easy navigation for tasks, requests, approvals, and tracking.

2. Employee and Admin Readiness

Both employee and administrative users are ready and capable of adopting the new system during the pilot phase. Clear instructions, support, and onboarding sessions will ensure smooth adoption.

3. Workflow Fit

The system fits well into current workflows by digitizing existing manual processes (like task assignments, leave requests, or reporting) without requiring major operational changes.

4. Training and Support Availability

Basic training is easy to deliver and understand. In addition, ongoing technical support will be provided during the pilot to resolve any user or system issues promptly.

5. Smooth Integration in Daily Use

The system does not disrupt daily operations. It enhances efficiency and communication, making everyday tasks more manageable for both employees and admin staff.

6. Feedback and Improvement Loop

Employees and admins can provide feedback during the pilot, allowing the system to be adjusted to meet real operational needs before full deployment.

CHAPTER 4: PROJECT DESCRIPTION

4.1: MODULES OF PROJECT:

The Project Pilot for Project Management System is a trial implementation of a digital platform designed to improve how projects are planned, managed, tracked, and delivered within an organization. This pilot will involve selected teams, both employees and administrators, to evaluate the system's performance and usability before launching it on a larger scale.

The system includes core project management features like task creation, scheduling, team collaboration, real-time progress tracking, and reporting. Admins can create projects, assign tasks, and monitor progress, while employees can view their assignments, update task status, track time, and communicate with their team.

- **Modules for Admin**

1. User Management

- Add/edit/delete employee accounts.
- Assign user roles (admin, manager, employee).
- Manage login access and passwords.

2. Project Management

- Create and assign projects.
- Define project goals, timelines, and budgets.
- Assign project managers and teams.

3.Task Oversight

- Monitor all tasks and team progress.
- Reassign tasks and change deadlines if needed.
- Set task priorities.
- Reports & Analytics.

4.View overall project status reports

- Track employee performance and productivity.
- Generate downloadable reports.

5.Feedback Collection

- Send feedback forms to employees.
- View feedback and suggestions for improvement.

6. System Settings

- Manage system-wide configurations.
- Control notification settings.
- Set up permissions and roles.

➢ **Modules for Employee/User**

1. Dashboard

- View assigned projects and tasks.
- Notifications and deadlines.
- Quick overview of personal performance.

2. Task Management

- View assigned tasks.
- Mark tasks as started/in-progress/completed.
- Submit progress updates.

3. Time Tracking

- Log hours worked on tasks.
- View work history.

4. Collaboration

- Comment on tasks.
- Upload files related to assigned work.
- Chat with team members (if enabled).

5.Project View

- See project timelines, milestones, and goals.
- Understand how individual work contributes to the whole project.

CHAPTER 5: DATA FLOW DIAGRAM

DFD: Data Flow Diagram

Data Flow Diagrams (DFDs) were first developed by Larry Constantine as a method of expressing system requirements in a graphical form. Also known as bubble charts, DFDs aim to clarify system requirements and identify major transformations, ultimately aiding in the system design process.

DFD is a means of representing a system at any level of detail through a graphic network of symbols that illustrate data flows, data stores, data processes, and data sources/destinations.

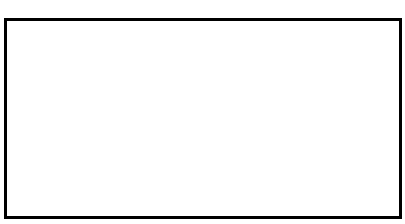
Purpose:

The primary purpose of data flow diagrams is to provide a semantic bridge between users and systems developers. They offer several key benefits:

- **Graphical Representation**: DFDs are graphical, which eliminates the need for lengthy textual descriptions and makes complex systems easier to understand.
- **Logical Representation**: They model what a system does, focusing on the logical aspect rather than the physical implementation. This helps in understanding the functionality without getting bogged down by technical details.
- **Hierarchical Structure**: DFDs are hierarchical, allowing systems to be shown at any level of detail. This makes it easier to break down complex processes into simpler, manageable components.
- **User Understanding and Review**: By providing a clear, visual representation of the system, DFDs facilitate user understanding and make it easier for users to review and provide feedback.

DFD Symbols are as follows:

- The External Entity symbol represents sources of data to the system or destinations of data from the system.



- The Data Flow symbol represents the movement of data.

- The Data Store symbol represents data that is not moving (delayed data at rest).

- The Process symbol represents an activity that transforms or manipulates the data.

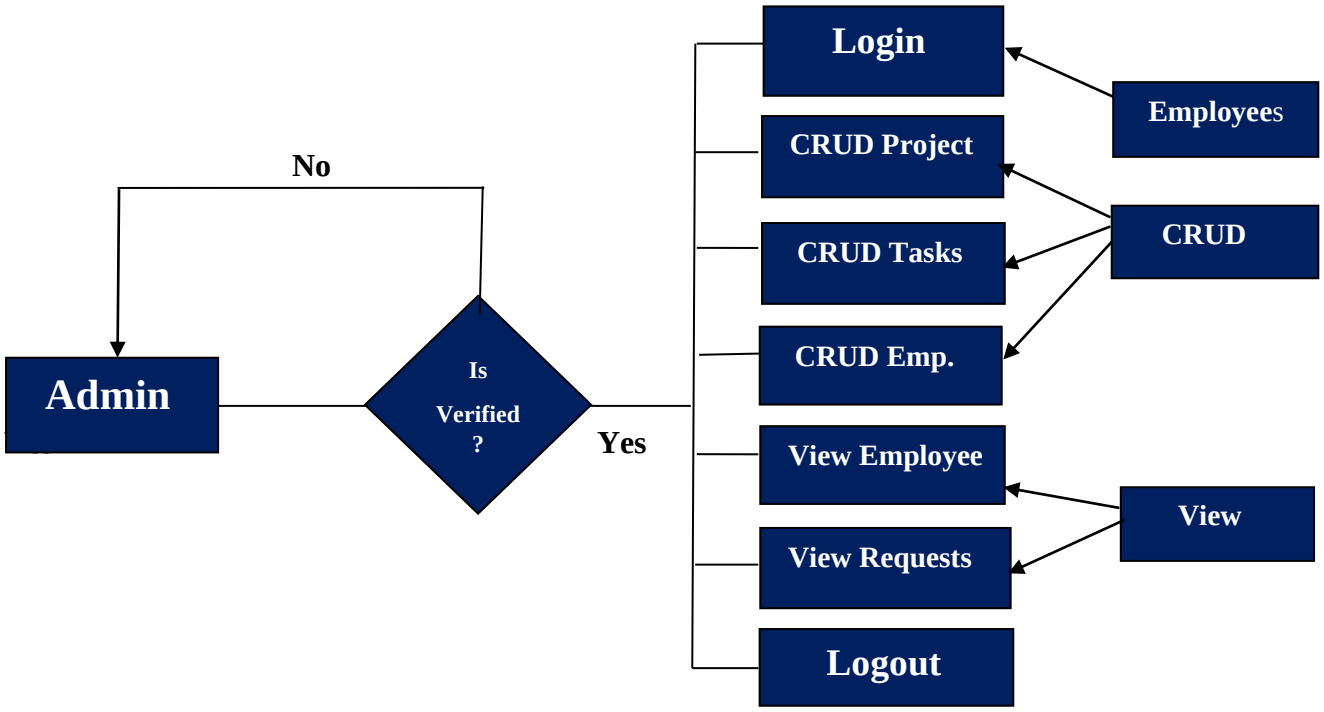
DFD Level 0

5.1: Context Level Diagram

Admin interacts with the "Project pilot" system to manage operations, while users engage with the system for browsing, booking, and ratings.

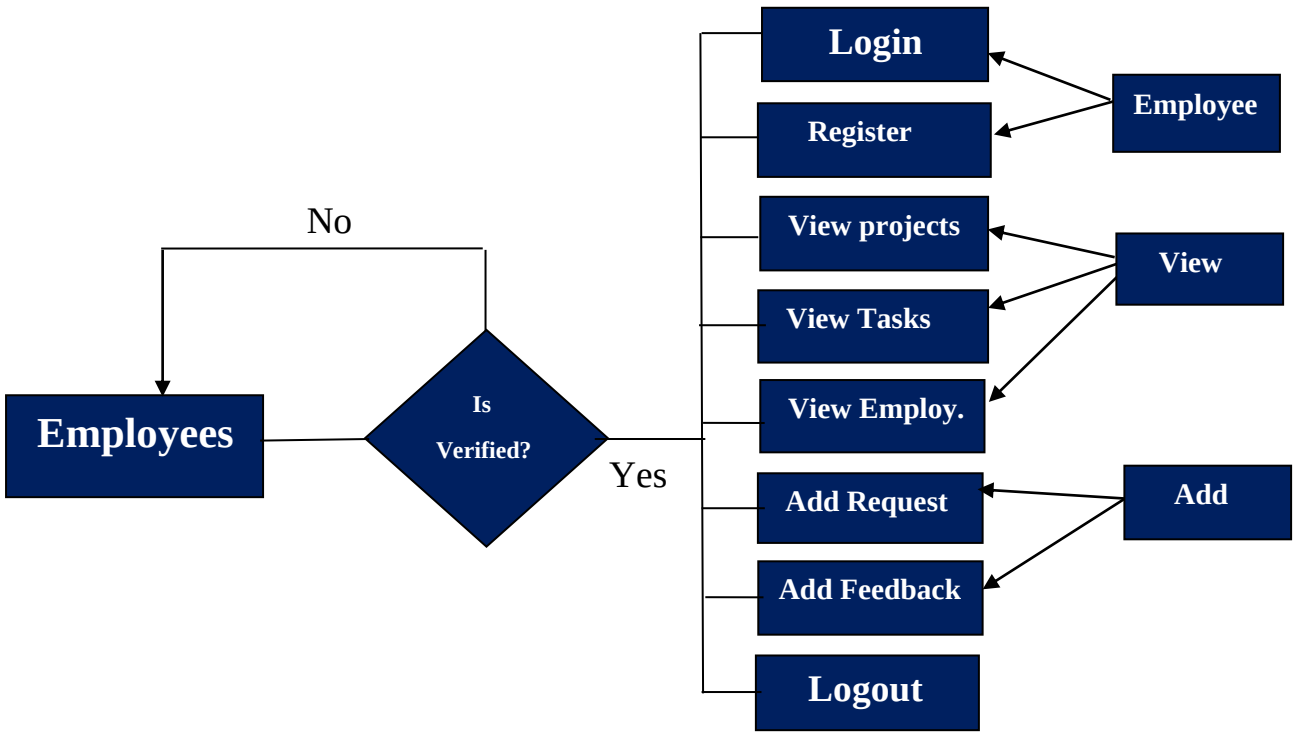
5.2: Level 1 DFD:

DFD for Admin

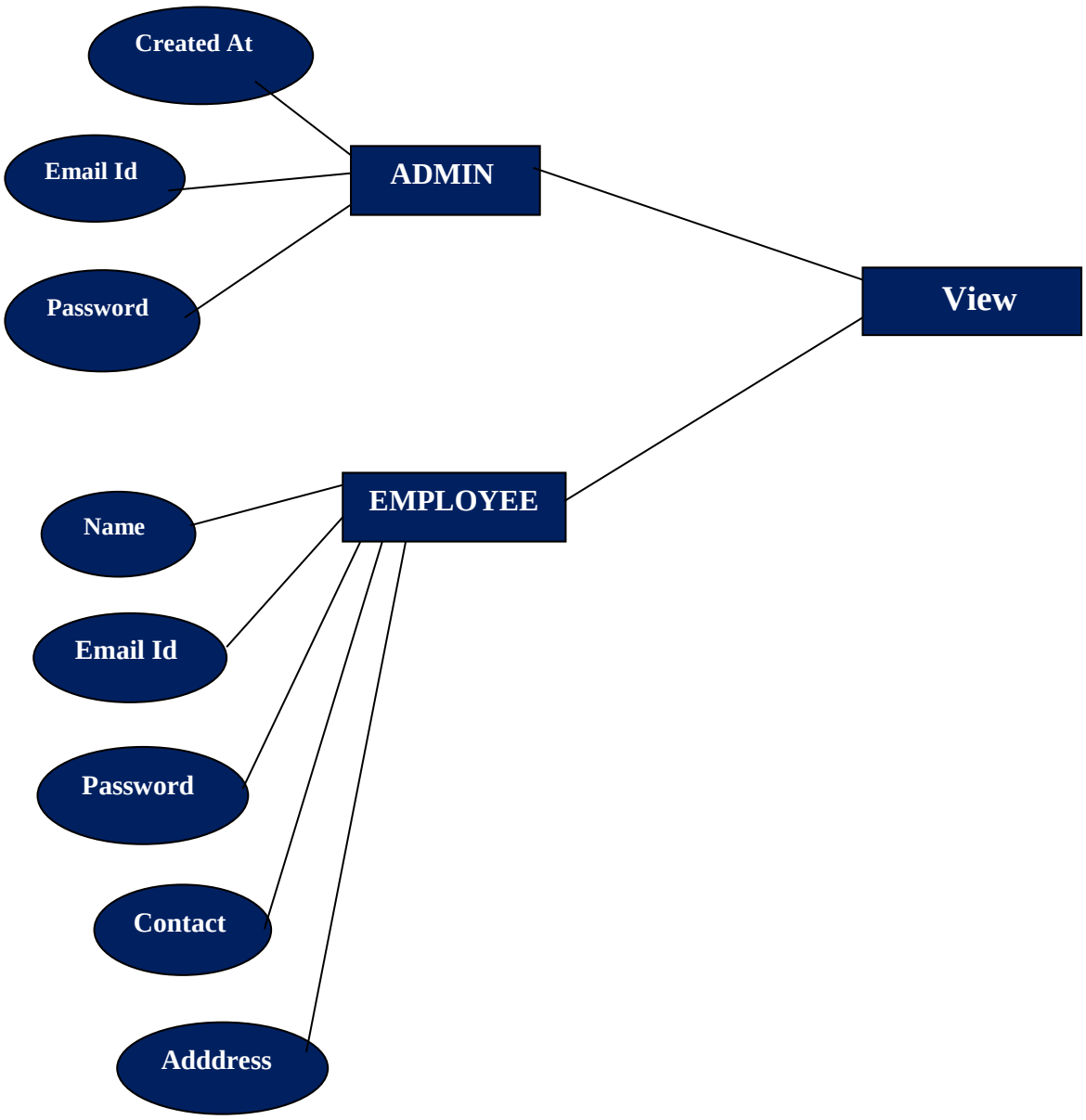


5.3: Level 2 DFD:

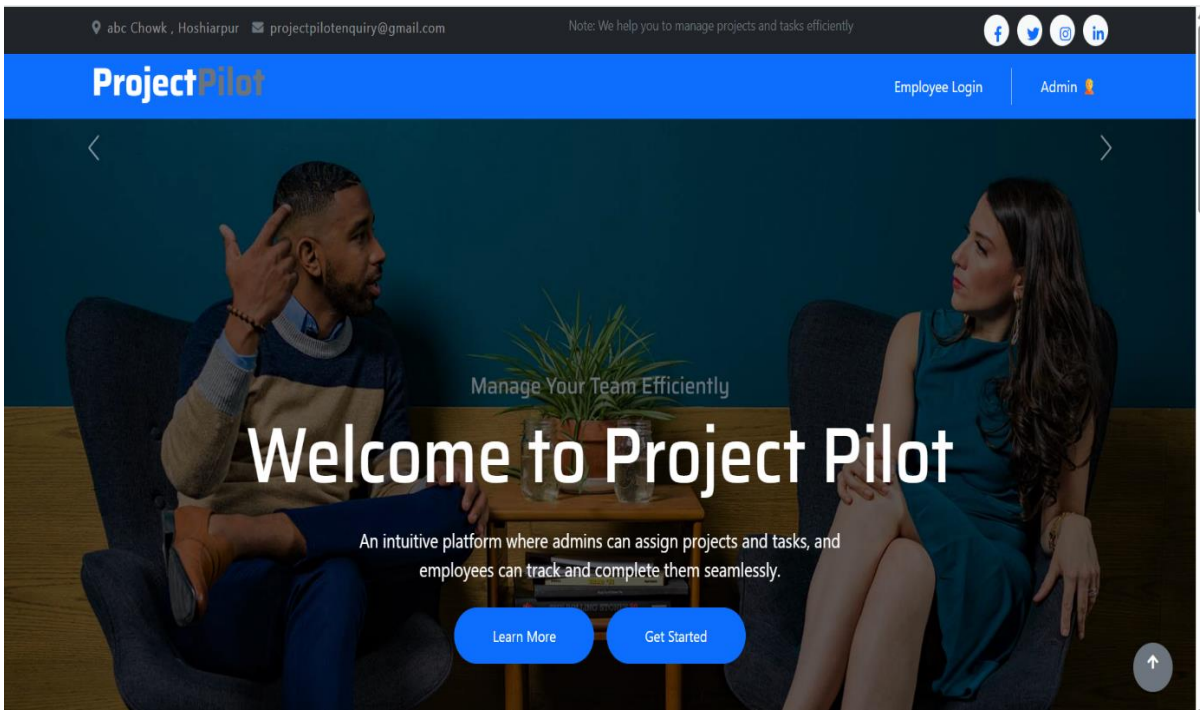
DFD for Employees

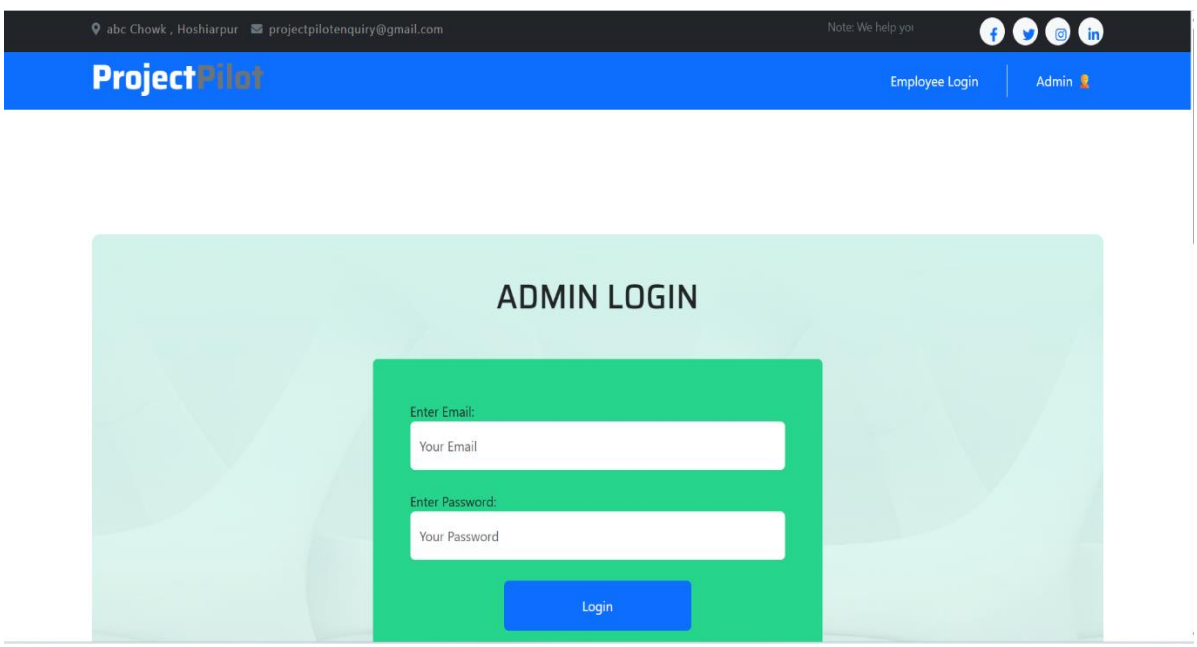


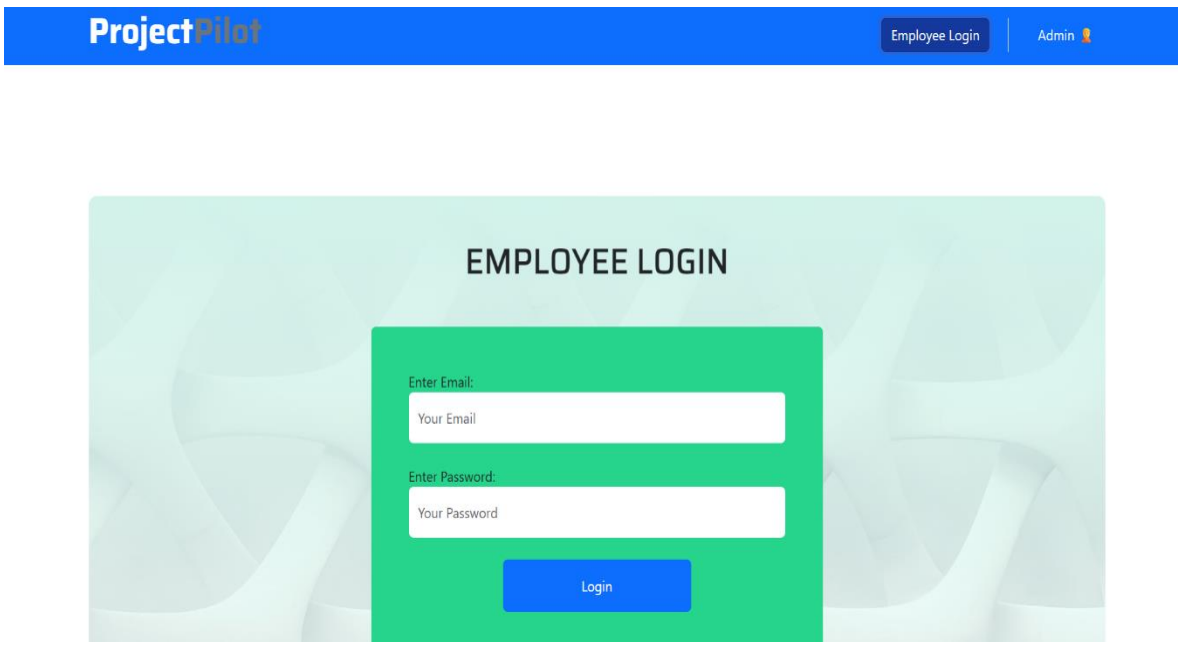
CHAPTER 6: ER DIAGRAM



CHAPTER 7: WORKING SCREENSHOT



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Admin
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Your Email
Enter Password:
Your Password
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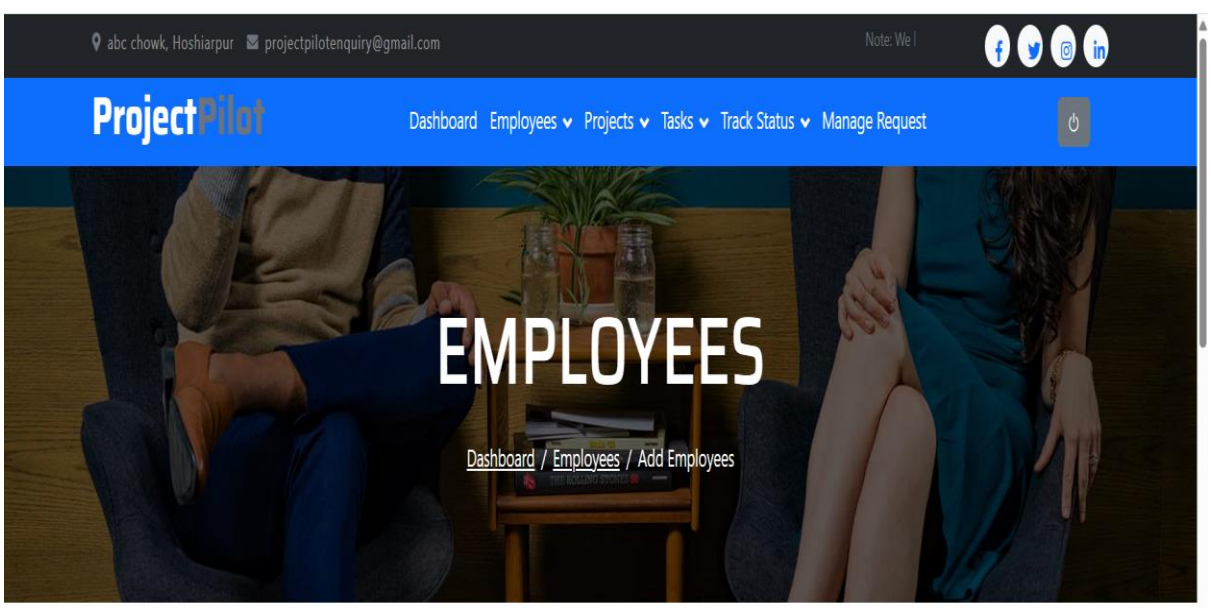

abc chowk, Hoshiarpur
projectpilotenquiry@gmail.com
ProjectPilot
Dashboard
Employees
Projects
Tasks
Track Status
Manage Request
EMPLOYEES
Dashboard / Employees / Add Employees

MANAGE EMPLOYEES

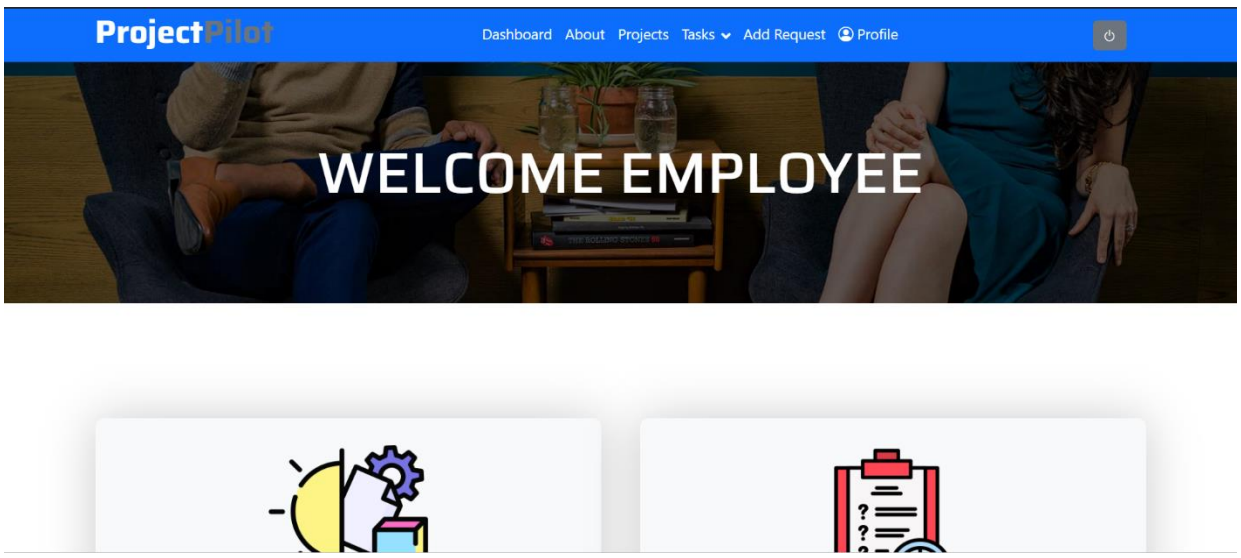
Sr No.	Name	Email	Image	Designation	Contact	Address	Status	Action
1	Priya Malhotra	priya@123gmail.com		Javascript developer	8435345344	Chandhigarh	Active	
2	Harsh Bhatia	officialharsh@company.com		Front-end developer	8798093832	Delhi	Active	
3	Alisha	alisha12@gmail.com		React Developer	7865786578	Hsp	Active	

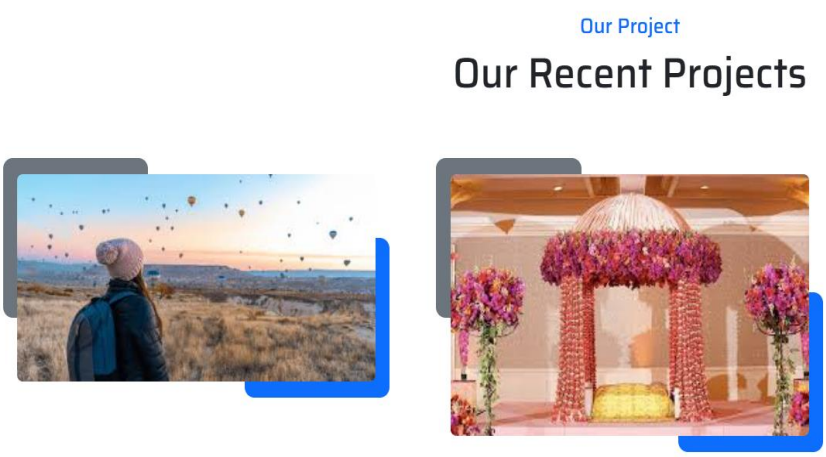
MANAGE PROJECTS

Sr No.	Name	Deadline	Image	Technology	Description	Status	Action
1	Travel 4U website	2025-08-06		Javascript	There is no description	Active	
2	Eventease website	2025-07-30		React JS & Firebase	No	Active	

MANAGE TASKS

Sr No.	Title	Deadline	Project Info	Employee Info	Priority	Description	Status	Action
1	project	no		Name: Email: Contact: Address:		no	Active	
2	Excel Sheet	2025-08-01		Name: Alisha Email: alisha12@gmail.com Contact: 7865786578 Address: Hsp	High	abc	Active	
3	Firebase database	2025-08-06	Travel 4U website	Name: Priya Malhotra Email: priya@123gmail.com Contact: 8435345344 Address: Chandhigarh	High	xyz	Active	
4	spread sheets	2025-07-30	Eventease website	Name: Harsh Bhatia Email: officialharsh@company.com Contact: 8798093832 Address: Delhi	Medium	xyz	Active	


ProjectPilot
Dashboard
About
Projects
Tasks
Add Request
Profile
WELCOME EMPLOYEE


Our Project
Our Recent Projects

CHAPTER 8 : CONCLUSION & APPLICATIONS

- **Tested Core Features:**

  The pilot helped in testing key functionalities like task assignment, employee data management, time tracking, and admin controls.

- **Identified Issues Early:**

  It allowed us to detect technical problems, user experience challenges, and areas that need improvement before full deployment.

- **Gained Valuable Feedback:**

  Feedback from users (employees/admins) provided insights into how intuitive and efficient the system is in real-world use.

- **Improved Workflow Visibility:**

  The system showed potential in improving overall workflow transparency, team collaboration, and productivity.

- **Validated System Feasibility:**

  The pilot proved that the system is technically, economically, and operationally feasible for the organization.

- **Enabled Continuous Improvement:**

  Based on the pilot results, necessary changes and enhancements can now be made to build a stronger final product.

APPLICATIONS:

- **Testing feasibility:**

  A pilot project helps determine if the core concept of the project is viable and can be practically implemented.

- **Risk mitigation:**

  By identifying potential problems early, a pilot project allows for adjustments and risk mitigation strategies to be developed before a full launch.

- **Resource allocation:**

  Pilot projects help in assessing the resource requirements (time, budget, personnel) for the full-scale project.

- **Stakeholder engagement:**

  Pilot projects offer opportunities to gather feedback from stakeholders and ensure buy-in for the larger project.

CHAPTER 9: FUTURE SCOPE

1. Enhanced Risk Management:

- Pilot projects will be utilized to identify potential risks and challenges in a controlled environment, allowing for adjustments before full-scale deployment.
- This includes assessing technical, operational, financial, and even cultural risks associated with new technologies or methodologies.
- By understanding and mitigating risks early, organizations can avoid costly mistakes and delays during the full project lifecycle.

2. Data-Driven Decision Making:

- Pilot projects will be designed to gather specific data and insights that inform project decisions.
- This data can be used to validate assumptions, refine strategies, and optimize processes based on real-world results.
- For example, pilot projects can help determine the scalability of a new system or the effectiveness of a new marketing campaign.

3. Learning and Skill Development:

- Pilot projects offer valuable learning opportunities for project teams and stakeholders.
- They allow teams to gain hands-on experience with new technologies, processes, or methodologies.
- This knowledge can then be applied to future projects, leading to improved efficiency and success rates.

4. Increased Focus on Specific Areas:

- Pilot projects will likely become more targeted, focusing on specific aspects of a larger project.
- For instance, a pilot project might focus solely on testing the integration of a new AI tool with existing systems, rather than attempting to implement the entire AI solution at once.
- This allows for more in-depth analysis and refinement of specific components before they are integrated into the larger project.

5. Broader Adoption of Agile Methodologies:

- Agile methodologies, which emphasize iterative development and continuous feedback, are well-suited for pilot projects.
- The flexibility of agile allows for adjustments and improvements based on the insights gained during the pilot phase.
- This approach can lead to more responsive and effective project outcomes.

REFRENCES AND LINKS

- HTML:-(https://www.w3schools.com/html)
- CSS:-(https://www.geeksforgeeks.org/css/css-tutorial)
- JAVASCRIPT:-(https://www.w3schools.com/js/)
- BOOTSTRAP:-(https://getbootstrap.com/)
- REACT JS:-(https://react.dev/)
- FIREBASE:-(https://firebase.google.com/)